

EXACT LOCAL RIGIDITY OF A RANK-23 DECOMPOSITION OF THE 3×3 MATRIX-MULTIPLICATION TENSOR OVER $\mathbb{F}_2$

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ABSTRACT. The exact tensor rank of 3×3 matrix multiplication remains open, including over $\mathbb{F}_2$, where the presently certified interval is $[20, 23]$. We study a frozen rank-23 decomposition obtained by reducing a public integral decomposition modulo two. For every subset of at most eight of its summands, we prove that the sum of those summands has rank equal to the size of the subset. Consequently, no replacement of at most eight published products by fewer products can lower the rank. The proof combines intrinsic mode supports with a slice-space extension criterion and an exhaustive exact computation over 880,946 subsets. All arithmetic is bit-exact; the input decomposition is independently checked against all 729 Brent identities. This is a local obstruction, not a proof that the full tensor has rank 23.

1. INTRODUCTION

Laderman exhibited a 23-multiplication algorithm for 3×3 matrix multiplication in 1976 [2]. Whether 22 multiplications suffice is still unknown. Over $\mathbb{F}_2$, Wang recently certified the lower bound 20 by proof-carrying backtracking [3]. Thus the exact binary rank is currently known only to lie between 20 and 23. Karunaratne and Idamekorala gave an integral rank-23 realization with 55 additions and exact checks of all Brent identities [1].

This note records a rigorous obstruction to a natural upper-bound strategy: remove a small group of products from that rank-23 decomposition and replace it by fewer arbitrary products. The replacement factors are not restricted to the published supports or coefficient patterns. Our endpoint is the following finite theorem.

Theorem 1.1 (Local rigidity). *Let*

$$M_{\langle 3,3,3 \rangle} = t_1 + \cdots + t_{23}$$

be the rank-23 decomposition over $\mathbb{F}_2$ specified by the frozen certificate with SHA-256

91c36b1165a71d4ba78cd8eb9c51ef9666d05e9162fdef5cba28d1d697a17dffa.

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Then for every $I \subseteq \{1, \dots, 23\}$ with $2 \leq |I| \leq 8$,

$$\mathbf{R}_{\mathbb{F}_2} \left(\sum_{i \in I} t_i \right) = |I|.$$

The theorem excludes 880,946 local replacement problems. It does not compare the published decomposition with an unrelated rank-22 decomposition and therefore does not improve the global lower bound.

2. DEFINITIONS

Let U, V, W be finite-dimensional vector spaces over $\mathbb{F}_2$. A pure tensor is a nonzero tensor $u \otimes v \otimes w$. The rank $\mathbf{R}(T)$ of $T \in U \otimes V \otimes W$ is the least number of pure tensors whose sum is T . For the matrix-multiplication tensor we use the coordinate convention

$$M_{\langle 3,3,3 \rangle} = \sum_{i,j,k=0}^2 E_{ij}^* \otimes E_{jk}^* \otimes E_{ik}.$$

Equivalently, a length- R factorization with coefficient arrays $\alpha_{s,ij}$, $\beta_{s,jk}$ and $\gamma_{ik,s}$ satisfies the $3^6 = 729$ Brent identities

$$\sum_{s=1}^R \alpha_{s,ij} \beta_{s,jk} \gamma_{ik,s} = \delta_{im} \delta_{jk} \delta_{ln}.$$

Fix a mode, say U . Contraction defines a linear map

$$T_U : U^* \longrightarrow V \otimes W.$$

Its image $S_U(T)$ is the U -slice space, and its dimension is the U -mode rank. The three intrinsic mode supports are the column spaces of the three flattenings. Projecting an arbitrary decomposition onto these supports never increases its length.

3. A SLICE-SPACE EXTENSION CRITERION

The following elementary criterion is the main reduction used by the certificate.

Lemma 3.1 (Pure extension criterion). *Let $S = S_U(T)$ have dimension r , and let $d \geq 0$. Then $\mathbf{R}(T) \leq r + d$ if and only if there is a subspace $Q \subseteq V \otimes W$ such that*

- (1) $S \subseteq Q$;
- (2) $\dim Q \leq r + d$; and
- (3) Q has a basis consisting of rank-one matrices $v \otimes w$.

It is enough to search for Q inside the tensor product of the intrinsic V - and W -supports of T .

Proof. Suppose $T = \sum_{a=1}^q u_a \otimes v_a \otimes w_a$ with $q \leq r + d$. Every slice belongs to $Q = \text{span}\{v_a \otimes w_a : 1 \leq a \leq q\}$. After discarding dependent generators, Q has a basis of rank-one matrices and $\dim Q \leq q$. Projection onto the intrinsic supports fixes T and maps every summand to a pure tensor or zero, proving the final assertion.

Conversely, take a rank-one basis $q_1, \dots, q_e$ of Q , where $e \leq r + d$. Express every U -slice in this basis. The coordinate functionals assemble into vectors $u_1, \dots, u_e \in U$, and the tensor-map identification gives $T = \sum_{a=1}^e u_a \otimes q_a$. Zero terms may be discarded, so $R(T) \leq e$. $\square$

For $d = 0$, one tests whether the rank-one elements of S span S . For $d = 1$, every candidate extension is $S + \langle q \rangle$ for a rank-one q . For $d = 2$, it suffices to enumerate two rank-one cosets in $(V \otimes W)/S$. Over $\mathbb{F}_2$, intrinsic supports of dimensions at most six make these enumerations small: a pair of six-dimensional supports contains only $(2^6 - 1)^2 = 3969$ nonzero rank-one matrices.

4. EXACT FINITE CERTIFICATE

The input arrays have shapes 23×9 , 23×9 , and 9×23 , with coefficients in $\{-1, 0, 1\}$. They are reduced modulo two and converted to 729-bit pure tensors. Before the local search, a separate verifier checks all 729 Brent identities both over $\mathbb{Z}$ and over $\mathbb{F}_2$. No summand becomes zero and no two summands coincide after reduction.

For each subset I , let $T_I = \sum_{i \in I} t_i$. The upper bound $R(T_I) \leq |I|$ is immediate. To exclude rank $|I| - 1$, the verifier first computes the three flattening ranks exactly. For $|I| \leq 4$, it enumerates every pure tensor in the intrinsic supports and tests all sums of at most $|I| - 1$ pure tensors. For $|I| = 5$, it uses Lemma 3.1 with $d = 0$ after the flattening test. For $|I| = 6, 7, 8$, it uses the same lemma with $d = 0, 1, 2$ as required by the mode-rank deficit. Any case outside the implemented criteria is counted as unresolved rather than rejected. The final run has zero unresolved cases.

TABLE 1. Exact local replacement audit.

k	subsets $\binom{23}{k}$	compressions	unresolved
2	253	0	0
3	1,771	0	0
4	8,855	0	0
5	33,649	0	0
6	100,947	0	0
7	245,157	0	0
8	490,314	0	0
total	880,946	0	0

The decisive run took 390.0 seconds on an Apple M3 Max and produced a JSON record with SHA-256

9000353f39cae386cde21006e7eed57b1f5dae99cb32f20d14995d9224012c0d.

The verifier source used for that run has SHA-256

b8a67ca1b57feb6285eb8eb96364351d94b497a3a9267e4d9d3af6c095240a0f.

Proof of Theorem 1.1. For every listed subset, the exact audit proves

$$R(T_I) > |I| - 1.$$

The published summands give $R(T_I) \leq |I|$. Thus equality holds. $\square$

Corollary 4.1. *No decomposition obtained from the frozen rank-23 decomposition by deleting at most eight summands and inserting fewer arbitrary pure tensors has rank below 23.*

Proof. A replacement of k deleted terms by at most $k - 1$ terms would give rank at most $k - 1$ for their sum, contradicting Theorem 1.1. $\square$

5. REPRODUCED GLOBAL BOUNDS AND FAILED ATTACKS

The official lower-bound certificate for $R_{\mathbb{F}_2}(M_{\langle 3,3,3 \rangle}) \geq 20$ was downloaded through Git LFS and checked against its two recorded hashes. The independently built C++ verifier replayed all restriction orbits in 3.50 seconds and reported the unconstrained lower bound 20.

For the upper bound, five fixed-seed flip-graph searches performed a total of 336 million local updates. Every run stopped at rank 23; each output was independently parsed and checked against all 729 identities. The five additional decompositions were also locally audited through five-term replacements, with no compression.

For the lower target 21, the existing orbit table was held fixed and the unconstrained backtracking endpoint was rerun with budgets 10^8 and 10^9 DFS calls. The larger run took 258.2 seconds, used approximately 42.8 GB peak resident memory, exhausted its budget, and left the lower-bound certificate unchanged at 20. This is negative search evidence only.

6. LIMITATIONS AND NEXT STEPS

Theorem 1.1 is a theorem about the neighborhood of one decomposition. It does not imply that an unrelated rank-22 decomposition is impossible. Likewise, failure of the current proof-carrying backtracking rules at target 21 does not imply that the lower bound is false.

The most concrete next lower-bound task is to strengthen the bounds for selected constrained orbits before rerunning the unconstrained endpoint. The most concrete upper-bound task is an unrestricted rank-22 XOR-SAT encoding with symmetry breaking and a model-to-Brent certificate path. Any SAT model is short and independently checkable; an UNSAT conclusion requires a DRAT/LRAT-style proof or an equivalent exact finite certificate.

DATA, CODE, AND COMPUTATIONAL DISCLOSURE

The project directory contains the frozen inputs, exact verifiers, complete JSON records, fixed seeds, build logs, and hash manifest. Python 3.9.6, Z3 5.1.0, Bazelisk 1.29.0/Bazel 8.3.1, Apple Clang 16.0.0, and Node.js 22.21.0 were used on macOS 26.6.1. AI assistance was used for proof exploration, implementation, testing, literature retrieval, and drafting. All stated mathematical endpoints are supported by exact arithmetic or a proof-carrying verifier; heuristic failures are identified explicitly.

REFERENCES

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2. Julian D. Laderman, *A noncommutative algorithm for multiplying 3×3 matrices using 23 multiplications*, Bulletin of the American Mathematical Society **82** (1976), no. 1, 126–128.
3. Chengu Wang, *Automated lower bounds for bilinear complexity over finite fields*, 2026.

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